

Mahmoud Hosny

Software Engineer · Machine Learning & AI · Data Analysis

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PROFILE

Computer Science student at Alexandria University with hands-on experience across the full development spectrum — from object-oriented system design in Java and C++ to building and evaluating machine learning models in Python, R, and SAS. Proficient in the complete ML pipeline: data wrangling, feature engineering, model training, cross-validation, and evaluation. Demonstrated ability to deliver 12+ academic and personal projects spanning ML, data analysis, backend systems, and frontend development. Strong communicator with a collaborative mindset and a commitment to clean, correct, and impactful software.

TECHNICAL SKILLS

Languages	Java, Python, C++, R, SQL, SAS, JavaScript, HTML, CSS3, SQLite
ML & Data	Linear Regression, KNN, Naïve Bayes, Classification, Gradient Descent, Cross-Validation, Feature Engineering, Data Mining, EDA
Tools	Git, R Studio, Java Swing, OOP Design, A* Search Algorithm, Problem Solving, Teamwork, Adaptability

PROJECTS

California House Pricing — Python · Regression · Statistics

- Conducted a comprehensive regression study on housing prices, comparing statistical, gradient descent, and numerical methods across 3+ model configurations.
- Produced visualizations and a comparative evaluation report highlighting each model's performance — achieving the lowest RMSE with gradient descent tuning.

FIFA Players Performance Analysis — Python · ML · Feature Engineering

- Built regression and classification models on a dataset of 18,000+ FIFA players to predict market value and classify performance tiers.
- Applied feature engineering and cross-validation, improving prediction accuracy by selecting the top 15 most impactful player attributes.

Car Price Prediction — Python · ML · Model Selection

- Compared Linear Regression, KNN, and Naïve Bayes models for predicting used car prices across a dataset of 10,000+ listings.
- Used cross-validation and metric-based evaluation to select the best-performing model, reducing prediction error by over 20% vs. baseline.

E-Commerce Churn Prediction — SAS · Classification · Logistic Modeling

- Predicted customer churn for an e-commerce platform using SAS classification models, identifying at-risk customer segments through logistic modeling.
- Achieved strong precision/recall balance through data preprocessing and feature selection on a structured transactional dataset.

City Supermarkets Analysis — R Studio · Statistics · EDA

- Performed exploratory and statistical analysis on supermarket sales data to uncover regional trends, customer behavior patterns, and top-selling categories.
- Generated 10+ visual reports to communicate findings clearly to a non-technical audience.

Student Grade Prediction — Python · Data Mining · Classification

- Applied data mining techniques to predict student academic performance, evaluating multiple classification models (Decision Tree, KNN, Logistic Regression).
- Performed feature selection to identify the 5 most predictive academic and behavioral attributes.

3×3 Puzzle Game with AI Solver — *Python · A* Algorithm*

- Built an interactive sliding puzzle with an AI solver using the A* search algorithm, finding optimal solutions via heuristic pathfinding in real time.

Library Administration System — *Java · OOP*

- Designed a full-featured library management system handling books, members, borrowing, and returns using clean object-oriented architecture.

Airplane BnB — Desktop App — *Java · Java Swing · SQLite*

- Developed a desktop booking application for airplane seats with a Swing GUI and SQLite persistence for 500+ simulated users and reservations.

E-Commerce Website Backend — *Java · OOP*

- Built the backend for an e-commerce platform covering product listings, order management, and user accounts with a layered OOP architecture.

University Registration System — *Java · OOP*

- Built an academic registration system managing student enrollment, course scheduling, and grade tracking with a clean layered OOP design.

Hotel Page — Frontend — *HTML · CSS3 · JavaScript*

- Designed and built a responsive hotel landing page with CSS3 animations and vanilla JavaScript for interactive UI components.

EDUCATION

Bachelor of Science — Computer Science / Software Engineering

2024 — 2028

Alexandria University, Egypt